

KW007 Radar Sensor GUI User Guide

Version: 2.6 **Date:** 2026-08-11

1. Introduction

The KW007 GUI application is a configuration and monitoring tool for the KW007 radar sensor module. This software allows you to:

- Configure sensor detection parameters
- Monitor real-time detection status

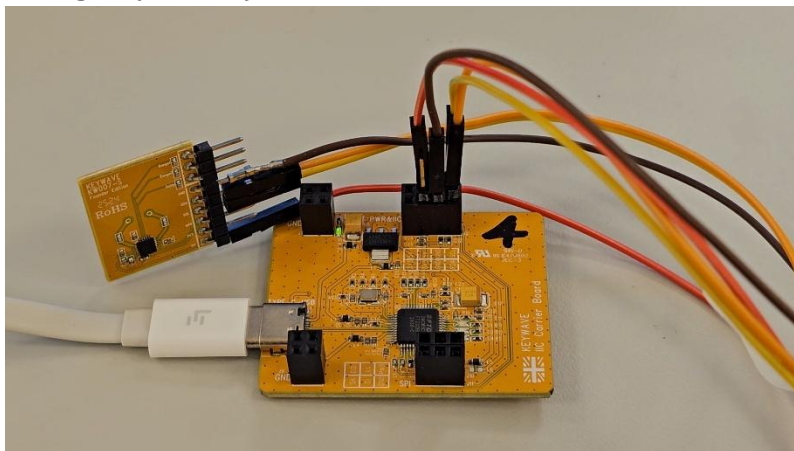
2. System Requirements

- **Operating System:** Windows 10 or later
- **Hardware:** FTDI USB-to-I2C adapter
- **Driver:** FTDI D2XX driver installed
- **Connection:** KW007 module connected via I2C

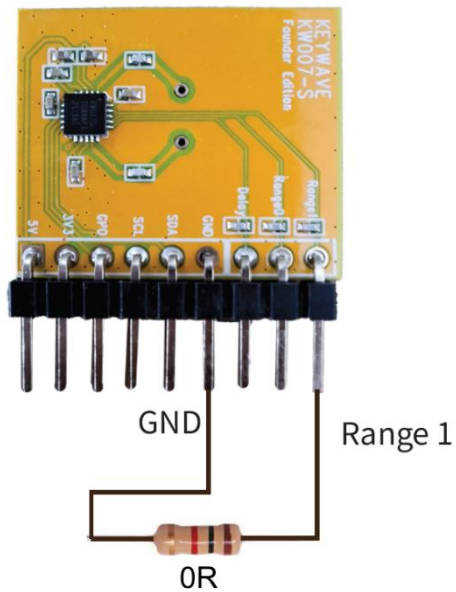
3. Getting Started

3.1 Connecting the Hardware

1. Connect the FTDI USB adapter to your computer
2. Connect the KW007 module to the I2C pins:
 - SDA (Data)
 - SCL (Clock)
 - VCC (Power)
 - GND (Ground)
 - Range1 (Ground)



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3.2 Hardware Note

When using KW007 in **I²C mode**, the IC pin 19 (Range1) must connect to GND.

3.3 Launching the Application

1. Double-click KW007_DemoUpDn_v2.6.exe to start the application
2. The main window will appear with all controls

4. Main Window Overview



*USB-to-I2C adapter : FT232H or CH341.

*Distance Range: Range1 is shortest, and Range8 is longest.

5. Custom Adjust Settings (Right side of GUI)

User can custom adjust setting after set-initial or fine-tune after set distance range on step2.

The detection range is determined by the three-stage amplifier gain (ampX_gain), the decision magnitude threshold (mag_th), Increasing the amplifier gain and reducing the magnitude threshold extends the detection range; however, these settings also increase susceptibility to false triggers.

Three-stage amplifier gain:

amp1_gain: (000=8dB, 001=12dB, 010=16dB, 011=19dB, 100=21dB, 101=24dB, 110=28dB, 111=32dB)

amp2_gain: (000=8dB, 001=12dB, 010=16dB, 011=19dB, 100=21dB, 101=24dB, 110=28dB, 111=32dB)

amp3_gain: (000=0dB, 001=4dB, 010=8dB, 011=12dB)

For example, to set gain to 48 dB:

Write amp1_gain 32dB, write amp2_gain 16dB, write amp3_gain 0dB.

Total gain:

$$32 + 16 + 0 = 48 \text{ dB.}$$

(We suggested to use higher gain in previous stage for better signal/noise quality)

Mag Threshold:

Set the signal threshold after three-stage amplifier, signal need larger than mag threshold to do detection algorithm. (Range 0~255).

Note: If gain too small or too high, the signal strength saturated to 0 or 255 and algorithm not work well.

Enhance Range Mode:

When check on, the detection range will increase.

The detection object speed is determined by doppler filter corner and sampling rate

Doppler Filter Corner:

This section configures the Doppler filter corner frequencies. The Doppler low-pass corner filters out fast-moving clutter, but may also attenuate fast-walking human targets. The Doppler high-pass corner is related to slow-motion clutter rejection.

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Sampling Rate:

Sampling rate will influence filter corner. Double sampling frequency will also double filter corner frequency.

Ex. $\text{Corner}_{fs}(1\text{kHz}) = 2 \times \text{Corner}_{fs}(500\text{Hz})$.

LightFlagDelay:

Defines the hold time for the GPO output. To prevent the output from turning ON and OFF too quickly (chattering) when a target is at the edge of the detection range. Once motion is detected, the GPO triggers. When the motion stops, the GPO will wait for the Delay Time to expire before returning to the "Idle" state.

Detect Period:

I2C report rate, the shortest period is 128ms.